
Earbuds with attitude (Source: Nuheara)

patches find applications in sports, enterprise and industrial markets as well. It forecasts body sensor shipments to increase to 68 million in 2021 from 2.7 million units in 2015.

Some of the growing companies in the wearable patch space are Smartpatch, Kenzen and Lief Therapeutics.

Kenzen's Echo smart patch, for instance, collects data from sweat. Sweat contains electrolytes, metabolites, small molecules and proteins, which are measured by sensors on the patch. The data is analysed on a smartphone, and used to predict and alert users about conditions like increased glucose levels, dehydration or cramps.

According to Dr Sonia Sousa, co-founder and CEO of Kenzen, "By putting on a patch, people can get real-time data on dehydration level, heart rate, respiration and core body temperature to make decisions about their health."

Lief Therapeutics has developed a bio-sensing patch that sticks to the chest and tracks heart rate and breathing using an electrocardiography sensor. When it detects a stressful condition, the patch vibrates to help the user regain a calmer breathing rhythm. Apparently, Lief developed this meditation system after measuring and studying the heartbeats of monks in India!

## Clothe yourself in technology

You can now buy clothes with technology inside—either stitched or woven into the fabric. Smart clothes are used to study sweat, monitor gait, track activity and more. Some clothes also help to cool or warm your body according to environmental and health conditions.

Spire and Swim.com have joined hands to launch smart swim suits that automatically track the wearers' pool time. Spire's Health Tags, which are fitted near the beltline of the suit, automatically sync data to users' phone. Swim.com's app is used to make sense of the data.

Google and Levi are working on Project Jacquard, where the technology is actually woven in your clothing. So your dress becomes like a touch screen, and you can actually raise the volume by just putting your hand into your pocket, or take a call by tugging the collar!

Several famous designers are also beginning to understand that a bit of technology woven into a dress makes it futuristic and fashionable. Maddy Maxey of Loomia and Billie Whitehouse of Wearable X are some well-known names working on smart fabrics using conductive inks, built-in haptics and materials that react to their environment. In a media interview, Maxey mentioned that though the technology is developing well, the manufacturing processes that can take the products to the market are still missing.

## Heard of hearables?

Another emerging area of wearables is the so-called 'hearable.' This part of the industry, according to trend-watchers, has boomed in 2017 and is slated to get smarter this year. Some are hearing aids with smart twists, while others are smart ear-buds that anyone can use.

Bragi, which developed the first in-ear computer in 2015, is partnering with Mimi Hearing Technologies on Project Ears, which aims to create an FDA-approved solution for hearing issues such as tinnitus. The solution revolves around the fact that each person has unique hearing requirements. The intelligent hearing amplifiers perform a scientific hearing test to measure your unique ear-print and program themeselves to optimise your hearing. These give relief from tinnitus with embedded masking sounds that help you relax and forget the ringing in your ears, while you continue to hear other sounds from your surroundings.

There are also two new models of IQbuds from Australian start-up Nuheara. IQbuds Boost stands out with its high level of personalisation, customisation and amplification. Like Bragi, Nuheara has introduced a new feature called Ear ID, which evaluates the user's hearing profile to create a more personalised hearing experience. Another model, LiveIQ, also includes noise cancellation.

Innomdle Lab, a start-up incubated by Samsung's C-Lab, has come up with Sgnl—a unique hearable that is shaped not like an ear-bud but as a smart strap worn on your wrist! It uses bone conduction technology to transmit sound from the band through your hand and fingers, into your ear. Basically, once you get a notification, you just place your finger on your ear and hear. It works well for phone calls and notifications. Users can listen to music too, but who would want to keep fingers on their ear for that long? A microphone in the band lets users talk back to the caller or give instructions to their digital assistant.

Jabra's Active 65T earbud is for active people. Its four microphones


Hear simply by placing your fingertips on your ear (Courtesy: Innomdle Lab)